

FreshRoute AI

Real-Time Accessibility Intelligence & Dynamic Routing

Project Submission Draft

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Project Summary

FreshRoute AI is an AI-powered, crowdsourced accessibility mapping and navigation platform that continuously detects and updates real-world accessibility conditions. It moves accessibility information away from a static map layer and turns it into a living, confidence-scored intelligence system that tells users not just whether a location is accessible in general, but whether a specific path is accessible right now.

1. Problem Statement

People with mobility difficulties often cannot reliably determine whether a route, sidewalk, building entrance, ramp, elevator, or pedestrian crossing is actually accessible. Existing accessibility information is frequently fragmented, static, incomplete, or outdated.

A route marked as accessible today may become unusable tomorrow because of construction, illegally parked vehicles, flooding, damaged infrastructure, blocked sidewalks, or non-functional elevators and ramps. This creates a major gap between knowing that a location is accessible and knowing whether it is accessible right now.

FreshRoute AI addresses this problem by creating a continuously updated accessibility intelligence layer for urban environments.

2. Proposed Solution

FreshRoute AI is an AI-powered, crowdsourced accessibility mapping and navigation platform that continuously detects and updates accessibility conditions in real-world environments. The platform draws on smartphone and dashcam imagery, user-submitted reports, computer vision, geospatial data, and historical and temporal information to identify accessibility features and obstacles, including:

- Wheelchair ramps, stairs, curbs, and accessible entrances
- Elevators and tactile paths
- Blocked sidewalks, parked vehicles, and construction barriers
- Flooded pathways and other temporary obstacles

This detected information is converted into a dynamic accessibility map. Users select their mobility requirements and receive routes based not only on distance, but on the current accessibility condition of each route. The system also explains why a route was recommended, allowing users to see the accessibility conditions and confidence behind each routing decision.

3. How It Works

Step 1 — Capture. Images or videos are collected from participating users, smartphones, dashcams, or other permitted sources.

Step 2 — AI Detection. A computer vision model identifies accessibility features and potential obstacles.

Step 3 — Geospatial Mapping. Each observation is associated with its geographic location and mapped to the relevant road or pedestrian-path segment.

Step 4 — Freshness & Confidence. Every accessibility observation receives a confidence and freshness level based on detection confidence, source reliability, number of confirmations, time since last verification, and conflicting observations.

Step 5 — Dynamic Accessibility Graph. The system converts the collected information into an accessibility-aware routing network.

Step 6 — Personalized Navigation. Users request routes according to their accessibility requirements and receive dynamically updated route options.

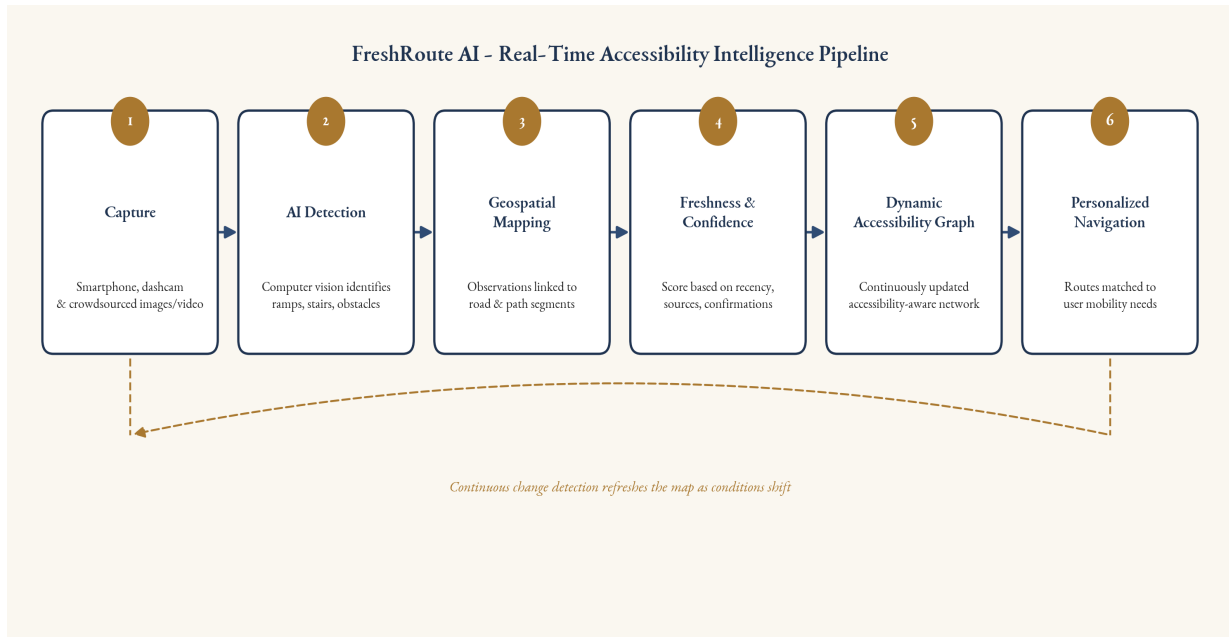


Figure 1. The FreshRoute AI real-time accessibility intelligence pipeline, from image capture through to personalized, accessibility-aware navigation. A continuous change-detection loop keeps the underlying map current.

4. Innovation / Novelty

FreshRoute AI transforms accessibility mapping from a static information layer into a continuously updated accessibility intelligence system. It combines computer vision, crowdsourced observations, temporal freshness, confidence estimation, and change detection to determine whether an accessibility condition is still valid and dynamically adapt routes accordingly.

Unlike conventional accessibility mapping systems that rely heavily on manually curated or periodically updated information, FreshRoute AI automates the detection of accessibility features and temporary obstacles while using temporal freshness and confidence scoring to continuously update route

accessibility.

Illustrative example. An obstacle observation may initially have a confidence score of 0.90. If no new observation confirms its presence for 72 hours, the proposed temporal-decay model may reduce its confidence to 0.41. A subsequent confirming observation can increase the confidence again. (These figures are illustrative of the proposed model, not measured results.)

1. Real-Time Accessibility Updates. The system detects temporary changes such as construction, flooding, blocked sidewalks, or inaccessible entrances and updates accessibility status accordingly.
2. Accessibility Freshness Score. Every piece of accessibility information carries a freshness indicator so users can see how recently a location was verified.
3. Confidence-Aware Accessibility. Reports are not treated as equally reliable; confidence is calculated from multiple observations and sources.
4. Change Detection. New observations are compared with previous ones to identify changes in accessibility conditions — for example, a sidewalk marked accessible yesterday may be automatically downgraded today if a construction barrier is detected.
5. Accessibility-Aware Dynamic Routing. The platform does not simply find the shortest route; it factors in accessibility constraints and current environmental conditions. The routing engine can trade off distance, accessibility, and information confidence — for example, selecting a slightly longer route with verified accessible infrastructure over a shorter route with uncertain or recently reported obstacles.

Detect → Verify → Decay → Update → Reroute

The core novelty in five words.

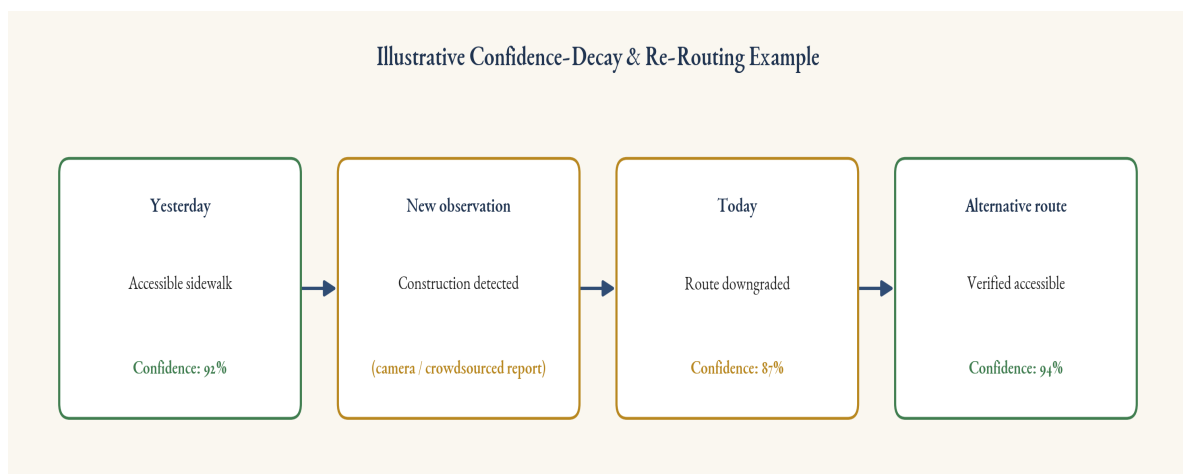


Figure 2. Illustrative confidence-decay and re-routing example: an accessible sidewalk loses confidence as it goes unconfirmed, a new observation detects a construction barrier, and the route is automatically downgraded in favor of a verified alternative.

5. Technology Stack

Layer	Technologies
Artificial Intelligence	Computer vision, object detection, image/video classification, change detection, confidence estimation, machine learning
Geospatial	GPS, OpenStreetMap, GIS, PostGIS, geospatial routing
Application	Flutter / React, web dashboard, mobile accessibility navigation interface
Backend	Python, FastAPI, PostgreSQL / PostGIS, REST APIs, near-real-time data processing
Optional	Edge AI for on-device detection, crowdsourced verification, cloud infrastructure, notification services

6. Expected Impact

FreshRoute AI can help wheelchair users, people with mobility impairments, elderly users, those with temporary injuries, parents using strollers, and other users who require accessible routes.

Social Impact. Reduces uncertainty and makes independent urban mobility safer and more predictable.

Smart City Impact. Cities can use aggregated accessibility data to identify recurring infrastructure problems and prioritize improvements.

Community Impact. User-contributed observations continuously improve accessibility information for everyone.

Global Impact. The concept can be deployed across cities and countries without a centralized accessibility database, building the accessibility layer progressively through community-generated data.

7. Illustrative Example

A wheelchair user wants to travel from Point A to Point B. A conventional map may identify the shortest pedestrian route. FreshRoute AI instead evaluates stairs, ramps, sidewalk condition, temporary obstacles, road crossings, elevator availability, recent reports, and the confidence and freshness of each observation. It may determine that the shortest route is currently blocked by construction and recommend an alternative route that is slightly longer but substantially more accessible.

The user can see why the route was selected and how recently the underlying accessibility information was verified.

8. MVP / Prototype Scope

For the initial prototype, FreshRoute AI will focus on a limited geographic area such as a college campus or selected urban neighborhood. The prototype will demonstrate:

- Image/video-based accessibility detection
- AI detection of obstacles and accessibility features
- Crowdsourced accessibility reports
- Dynamic accessibility map
- Freshness and confidence scores
- Accessibility-aware route generation
- User-facing mobile/web interface

The prototype can initially use simulated camera feeds and a curated dataset, running as a near-real-time prototype with an architecture designed for continuous real-world updates rather than live, city-wide infrastructure at this stage.

9. Scalability

FreshRoute AI is designed as a location-independent platform, capable of growing along a clear path:

Campus → Neighborhood → City → Multiple Cities → Global Accessibility Network

As more observations are collected, the system continuously improves its accessibility map and can provide additional data for future refinement of its detection models. The architecture can also support integration with municipalities, universities, transportation providers, accessibility organizations, and other permitted data sources.

10. Expected Outcome

FreshRoute AI shifts accessibility navigation from asking “Is this place accessible?” to answering three practical questions: “Is this path accessible now? How reliable is that information? What is the best accessible alternative?”

This creates a dynamic accessibility intelligence layer that can help make cities more inclusive, responsive, and navigable for everyone.